D-SCRAMBLED CANTOR SETS

B. MILLER

Abstract. For all $d \geq 3$, we provide an example of a homeomorphism of Cantor space that has an uncountable d-scrambled set but no d-scrambled Cantor set.

Introduction

Suppose that X is a metric space and $f \colon X \to X$. We say that $x \in X^d$ is *proximal* if $\liminf_{n \to \infty} \max_{i<j<d} d_X(f^n(x(i)), f^n(x(j))) = 0$, *asymptotic* if $\limsup_{n \to \infty} \min_{i<j<d} d_X(f^n(x(i)), f^n(x(j))) = 0$, and *Li–Yorke* if it is proximal but not asymptotic. A set $Y \subseteq X$ is *d-scrambled* if every injective sequence in Y^d is Li–Yorke, and f is *Li–Yorke d-chaotic* if there is an uncountable d-scrambled set $Y \subseteq X$.

In [GGM21], it was shown that every Li–Yorke 2-chaotic continuous function on a Polish space has a 2-scrambled Cantor set. Here we show that the analogous statement for $d \geq 3$ is false:

Theorem 1 ($\mathtt{ZFC}$)**.** *Suppose that $d \geq 3$. Then there is a Li–Yorke d-chaotic homeomorphism of $2^{\mathbb{N}}$ with no d-scrambled Cantor set.*

A *graph* on X is an irreflexive symmetric set $G \subseteq X \times X$. Let $\mathrm{Hyp}^d(G)$ be the d-ary relation on X of all injective sequences $x \in X^d$ such that $G \cap (x[d] \times x[d]) \neq \emptyset$. An *embedding* of a d-ary relation R on X into a d-ary relation S on Y is an injection $\pi \colon X \to Y$ such that $x \in R \iff \pi \circ x \in S$ for all $x \in X^d$. The *saturation* of a set $Y \subseteq X$ under a bijection $T \colon X \to X$ is given by $[Y]_T = \bigcup_{n \in \mathbb{Z}} T^n[Y]$. A subset of a topological space is F_σ if it is a countable union of closed sets. The primary new technical result underlying our proof of Theorem 1 is:

Theorem 2 ($\mathtt{ZF} + \mathtt{DC}$)**.** *Suppose that $d \geq 3$ and G is an F_σ graph on $2^{\mathbb{N}}$. Then there is a continuous embedding $\pi \colon 2^{\mathbb{N}} \to 2^{\mathbb{N}}$ of $\mathrm{Hyp}^d(G)$ into the d-ary Li–Yorkeness relation of a homeomorphism $T \colon 2^{\mathbb{N}} \to 2^{\mathbb{N}}$ for which the saturation of $\pi[2^{\mathbb{N}}]$ is co-countable.*

In §1, we establish Theorem 2. In §2, we combine this with known facts to establish Theorem 1 and a related independence phenomenon.

2010 *Mathematics Subject Classification.* Primary 54H20; Secondary 03E15.

Key words and phrases. Li–Yorke chaos, scrambled set, topological dynamics.

1. Embedding hypergraphs

For all $s \in 2^{<\mathbb{N}}$, define $\mathcal{N}_s = \{c \in 2^{\mathbb{N}} \mid \forall i < |s| \; s(i) = c(i)\}$. For each metric space X, define $T_X \colon X^{\mathbb{Z}} \to X^{\mathbb{Z}}$ by $T_X(x)(i) = x(i+1)$ for all $i \in \mathbb{Z}$ and $x \in X$. As Brouwer's theorem ensures that every non-empty compact perfect zero-dimensional metric space is homeomorphic to $2^{\mathbb{N}}$ (see, for example, [Kec95, Proposition 7.4]), Theorem 2 is a consequence of the following fact:

Theorem 1.1 ($\mathtt{ZF} + \mathtt{DC}$). *Suppose that $d \geq 3$ and G is an F_σ graph on $2^{\mathbb{N}}$. Then there exist a compact countable ultrametric space X, a T_X-invariant closed set $F \subseteq X^{\mathbb{Z}}$ whose finite sequences are proximal, and a continuous embedding $\pi \colon 2^{\mathbb{N}} \to X^{\mathbb{Z}}$ of $\mathit{Hyp}^d(G)$ into the complement of the d-ary asymptoticity relation for which the saturation of $\pi[2^{\mathbb{N}}]$ is a co-countable subset of F.*

Proof. Fix closed graphs G_n on $2^{\mathbb{N}}$ for which $G = \bigcup_{n \in \mathbb{N}} G_n$. For each $m \in \mathbb{N}$, let $G_{m,n}$ be the graph on 2^m consisting of all pairs of distinct sequences of the form $(c \restriction m, d \restriction m)$, where $(c, d) \in G_n$, and let $H_{m,n}$ be the $(d-1)$-ary relation on 2^m of all injective sequences $s \in (2^m)^{d-1}$ such that $s(0) \; G_{m,n} \; s(1)$. Fix an infinite set $I \subseteq \mathbb{N}$ such that $|I \cap (I - n)| < \aleph_0$ for all $n > 0$, as well as a bijection $\phi \colon I \to \mathbb{N} \cup \bigcup_{m,n \in \mathbb{N}} H_{m,n} \times \{n\}$.

Set $X = \{0, 1\} \cup (d \times \mathbb{N}) \cup \{\infty\}$, define $h_X \colon X \to \mathbb{R}$ by setting $h_X(0) = h_X(1) = 2$, $h_X(k, n) = 1/2^n$ for all $k < d$ and $n \in \mathbb{N}$, and $h_X(\infty) = 0$, let ρ_X be the compact ultrametric on X given by $\rho_X(x, y) = \max(h_X(x), h_X(y))$ for all distinct $x, y \in X$, and let ρ be the *product metric* on $X^{\mathbb{Z}}$ given by $\rho(x, y) = \sum_{i \in \mathbb{Z}} \rho_X(x(i), y(i))/2^{|i|}$.

Define $\pi \colon 2^{\mathbb{N}} \to X^{\mathbb{Z}}$ by setting $\pi(c)(i) = \infty$ whenever $i \in \mathbb{Z} \setminus I$ and

$$\pi(c)(i) = \begin{cases} c(\phi(i)) & \text{if } \phi(i) \in \mathbb{N}, \\ (k, \phi(i)(1)) & \text{if } k < d-1 \text{ and } c \in \mathcal{N}_{\phi(i)(0)(k)}, \text{ and} \\ (d-1, \phi(i)(1)) & \text{if } c \notin \bigcup_{k<d-1} \mathcal{N}_{\phi(i)(0)(k)} \end{cases}$$

whenever $i \in I$, for all $c \in 2^{\mathbb{N}}$. Clearly π is continuous.

Lemma 1.2. *The function π is injective.*

Proof. Suppose that $c, d \in 2^{\mathbb{N}}$ are distinct. Then there exists $n \in \mathbb{N}$ with $c(n) \neq d(n)$. Fix $i \in I$ for which $\phi(i) = n$. Then $\pi(b)(i) = b(n)$ for all $b \in \{c, d\}$, so $\pi(c)(i) \neq \pi(d)(i)$, thus $\pi(c) \neq \pi(d)$. □

A *homomorphism* from a d-ary relation R on Y to a d-ary relation S on Z is a function $\pi \colon Y \to Z$ such that $\pi \circ y \in S$ for all $y \in R$.

Lemma 1.3. *The function π is a homomorphism from $\mathit{Hyp}^d(G)$ to the complement of the d-ary asymptoticity relation.*

Proof. Suppose that $c \in \mathrm{Hyp}^d(G)$ and fix $j < k < d$ and $n \in \mathbb{N}$ for which $c \in \mathrm{Hyp}^d(G_n)$ and $c(j) \; G_n \; c(k)$. For all $m \in \mathbb{N}$, fix $s_m \in (2^m)^{d-1}$ such that $s_m(0) = c(j) \upharpoonright m$, $s_m(1) = c(k) \upharpoonright m$, and $c(\ell) \upharpoonright m \notin s_m[d-1]$ for at most one $\ell < d$. Then $I' = \phi^{-1}(\{(s_m, n) \mid m \in \mathbb{N}\})$ is infinite. But if $i' \in I'$, then $\{\pi(c(k))(i') \mid k < d\} = \{(k, n) \mid k < d\}$, so $\min_{j<k<d} \rho(T_X^{i'}(\pi(c(j))), T_X^{i}(\pi(c(k)))) \geq 1/2^n$. □

Lemma 1.4. *The function π is a homomorphism from $X^d \setminus Hyp^d(G)$ to the d-ary asymptoticity relation.*

Proof. Suppose that $c \in (2^{\mathbb{N}})^d$ and $\pi \circ c$ is not asymptotic. Fix $\epsilon > 0$ such that $I' = \{i \in \mathbb{N} \mid \min_{j<k<d} \rho(T_X^i(\pi(c(j))), T_X^i(\pi(c(k)))) \geq \epsilon\}$ is infinite, and $r \in \mathbb{N}$ with $\sum\{1/2^{|i|-1} \mid i \in \mathbb{Z} \setminus [-r, r]\} < \epsilon/2$. Our choice of I ensures that $I'' = \{i' \in I' \mid |I \cap [i'-r, i'+r]| \leq 1\}$ is co-finite in I', so our choice of r ensures that there is a unique $p(i'') \in I \cap [i'' - r, i'' + r]$ for all $i'' \in I''$. Then the set $I''' = p[I'']$ is infinite. Moreover, if $i'' \in I''$, then $\min_{j<k<d} \rho_X(\pi(c(j))(p(i'')), \pi(c(k))(p(i''))) \geq \epsilon/2$. As $d \geq 3$, it follows that $\phi(p(i'')) \notin \mathbb{N}$ and $\sup_{i''' \in I'''} \phi(i''')(1) < \infty$, so there exist an infinite set $I'''' \subseteq I'''$, distinct $j, k < d$, and $n \in \mathbb{N}$ such that $c(j) \in \mathcal{N}_{\phi(i'''')(0)(0)}$, $c(k) \in \mathcal{N}_{\phi(i'''')(0)(1)}$, and $n = \phi(i'''')(1)$ for all $i'''' \in I''''$. As G_n is closed and the lengths of $\phi(i'''')(0)(0)$ and $\phi(i'''')(0)(1)$ are unbounded, it follows that $c(j) \; G_n \; c(k)$, and therefore $c \in \mathrm{Hyp}^d(G_n)$, thus $c \in \mathrm{Hyp}^d(G)$. □

Let F be the closure of the T_X-saturation of $\pi[2^{\mathbb{N}}]$.

Lemma 1.5. *Suppose that $x \in F \setminus [\pi[2^{\mathbb{N}}]]_{T_X}$. Then $|\mathbb{Z} \setminus x^{-1}(\{\infty\})| \leq 1$.*

Proof. We will show that $|[-i, i] \setminus x^{-1}(\{\infty\})| \leq 1$ for all $i \in \mathbb{N}$. We can assume that there exist a sequence $(c_n)_{n \in \mathbb{N}}$ of elements of $2^{\mathbb{N}}$, $\epsilon \in \{\pm 1\}$, and a strictly increasing sequence $(i_n)_{n \in \mathbb{N}}$ of natural numbers such that $T_X^{\epsilon i_n}(\pi(c_n)) \to x$. Then the desired result follows from the fact that $|[\epsilon i_n - i, \epsilon i_n + i] \setminus \pi(c_n)^{-1}(\{\infty\})| \leq 1$ for all sufficiently large $n \in \mathbb{N}$. □

Lemma 1.5 ensures that the saturation of $\pi[2^{\mathbb{N}}]$ is co-countable in F. It also ensures that if $F' \subseteq F$ is finite, then $\bigcup\{\mathbb{Z} \setminus x^{-1}(\{\infty\}) \mid x \in F'\}$ is contained in a finite union of translates of I with a finite set. As the defining property of I ensures that such a set has arbitrarily long gaps, it follows that every finite sequence of elements of F is proximal. □

Remark. It is not difficult to modify the above argument so that it goes through for $G_{\delta\sigma}$ graphs while simultaneously ensuring that $X = 2$ and F is the one-point compactification of the saturation of $\pi[2^{\mathbb{N}}]$.

A set $Y \subseteq X$ is a *G-clique* if all distinct elements of Y are G-related, and *G-independent* if no elements of Y are G-related.

Proposition 1.6 ($\mathtt{ZF} + \mathtt{DC}$)**.** *Suppose that $d \geq 2$, X and Y are Polish spaces, G is a Borel graph on X, $T \colon Y \to Y$ is a Borel automorphism, $B \subseteq Y$ is an uncountable Borel d-scrambled set, $\pi \colon X \to Y$ is an injective Borel homomorphism from $X^d \setminus \mathit{Hyp}^d(G)$ to the complement of the d-ary Li–Yorkeness relation, and the saturation of $\pi[X]$ is co-countable. Then G has a perfect clique.*

Proof. Fix $i \in \mathbb{Z}$ for which $B \cap (T^i \circ \pi)[X]$ is uncountable. The perfect set theorem and Galvin's Theorem (see, for example, [Kec95, Theorems 13.6 and 19.7]) then yield a perfect set $P \subseteq (T^i \circ \pi)^{-1}(B)$ that is either a G-clique or G-independent. But G-independence would contradict the fact that B is d-scrambled. $\square$

2. D-Scrambled sets

We begin this section with the proof of our main result:

Proof (of Theorem 1). By [KV12, Theorem 2.1] and a simple Baire category argument, there is an F_σ graph G on $2^{\mathbb{N}}$ with an uncountable clique $Y \subseteq 2^{\mathbb{N}}$ but no perfect clique. By Theorem 2, there is a continuous embedding $\pi \colon 2^{\mathbb{N}} \to 2^{\mathbb{N}}$ of $\mathrm{Hyp}^d(G)$ into the d-ary Li–Yorkeness relation of a homeomorphism $T \colon 2^{\mathbb{N}} \to 2^{\mathbb{N}}$ for which the saturation of $\pi[2^{\mathbb{N}}]$ is co-countable. Then $\pi[Y]$ is an uncountable d-scrambled set and there is no perfect d-scrambled set by Proposition 1.6. $\square$

Let $\mathfrak{c}$ denote the cardinality of the continuum.

Theorem 2.1 ($\mathtt{ZFC} + \neg\mathtt{CH}$)**.** *Suppose that $d \geq 3$. Then it is consistent that there is a homeomorphism $T \colon 2^{\mathbb{N}} \to 2^{\mathbb{N}}$ with a d-scrambled set of cardinality $\mathfrak{c}$ but no perfect d-scrambled set.*

Proof. By [She99, Theorem 1.13], we can assume that there is an F_σ graph G on $2^{\mathbb{N}}$ with a clique of cardinality $\mathfrak{c}$ but no perfect clique. The proof of Theorem 1 therefore yields the desired result. $\square$

A *hypergraph* is a permutation-invariant set of injective sequences.

Theorem 2.2 ($\mathtt{ZFC} + \neg\mathtt{CH}$)**.** *Suppose that $d \geq 3$. Then it is consistent that every continuous function on a Polish space with a d-scrambled set of cardinality $\aleph_2$ has a d-scrambled perfect set.*

Proof. By [KS03, Proposition 3.4], we can assume that every Borel d-ary hypergraph on a Polish space with a clique of cardinality $\aleph_2$ has a perfect clique. But the d-ary Li–Yorkeness relation is Borel. $\square$

Remark. Given any set $D \subseteq \mathbb{N} \setminus 3$, the underlying ideas can be interleaved to obtain generalizations of our results to individual sets that are simultaneously d-scrambled for all $d \in D$.

Acknowledgements. I would like to thank Alexander Kechris for encouraging me to utilize ChatGPT-5.6 Sol, which provided substantial assistance in the discovery and development of the results.

References

[GGM21] Stefan Geschke, Jan Grebík, and Benjamin D. Miller, *Scrambled Cantor sets*, Proceedings of the American Mathematical Society **149** (2021), no. 10, 4461–4468.

[Kec95] Alexander S. Kechris, *Classical Descriptive Set Theory*, Graduate Texts in Mathematics, vol. 156, Springer-Verlag, New York, 1995.

[KS03] Wiesław Kubiś and Saharon Shelah, *Analytic colorings*, Annals of Pure and Applied Logic **121** (2003), no. 2–3, 145–161.

[KV12] Wiesław Kubiś and Benjamin Vejnar, *Covering an uncountable square by countably many continuous functions*, Proceedings of the American Mathematical Society **140** (2012), no. 12, 4359–4368.

[She99] Saharon Shelah, *Borel sets with large squares*, Fundamenta Mathematicae **159** (1999), no. 1, 1–50. MR 1669643

B. Miller, 1008 Balsawood Drive, Durham, NC 27705
Email address: `glimmeffros@gmail.com`
URL: `https://glimmeffros.github.io`